Assignment NO. 1 MTH501 (Fall 2021)

Maximum Marks: 15

Due Date: December 11, 2021

DON'T MISS THESE: Important instructions before attempting the solution of this assignment:

- **To solve this assignment, you should have good command over 1 - 7 lectures.**
- **Try to get the concepts, consolidate your concepts and ideas from these questions which you learn in the 1-7 lectures.**
- **Upload assignments properly through LMS, No Assignment will be accepted through email.**
- **Write your ID on the top of your solution file.**
- **Don't use colourful back grounds in your solution files.**
- **Use Math Type or Equation Editor Etc. for mathematical symbols.**
- **You should remember thatif we found the solution files of some students are same then we will reward zero marks to all those students.**
- **Try to make solution by yourself and protect your work from other students, otherwise you and the student who send same solution file as you will be given zero mark.**
- **Also remember that you are supposed to submit your assignment in Word format any other like scan images etc. will not be accepted and we will give zero mark corresponding to these assignments.**

Question 1: MARKS 5

For the following matrix A, show that $(p+q)A = pA + qA$
Take $p = 2$ and $q = 5$ as scalars.

$$A = \begin{bmatrix} 9 & 4 & 5 \\ 2 & 6 & 7 \end{bmatrix}$$

Question 2: MARKS 10

Solve the following system of equations by converting its augmented matrix to **reduced Echlon form**.

$$2x + y - z = 5$$
$$x - y + z = 1$$
$$x + 2y - 2z = 4$$